Spatial Focused Bitemporal Interactive Network for Remote Sensing Image Change Detection

Hang Sun, Yuan Yao, *Student Member, IEEE*, Lefei Zhang, *Senior Member, IEEE*, and Dong Ren

***Abstract*— Recently, transformers have been widely explored in remote sensing image change detection (RSICD) and achieved remarkable performance. However, most existing transformer-based change detection methods overlook exploring the spatiotemporal relationships between bitemporal images at the features within the same layer, which is crucial for learning discriminative features to perceive changes. Moreover, no explicit spatial constraint has been imposed on the final fused bitemporal features, leading to reduced detection performance on small targets. To address these issues, we propose a spatial focused bitemporal interactive network (SFBI-Net) for RSICD. Specifically, a bitemporal spatiotemporal interactive (BSI) module is proposed, which performs global interactions on bitemporal features at the same network layer and supplements local information to obtain spatiotemporal relationships of bitemporal features for discriminative representation. Furthermore, a spatial focus diversity loss (SFD-Loss) is developed to maximize bitemporal features in the spatial dimension and further enhance the feature representation of change areas, especially small target areas. The experimental results on challenging benchmark datasets demonstrate the superiority of our SFBI-Net. The source code is available at (https://github.com/Mryao-yuan/SFBI-Net).**

***Index Terms*— Bitemporal spatiotemporal interactive (BSI), remote sensing image change detection (RSICD), spatial feature focus constraint, transformers.**

I. Introduction

REMOTE sensing image change detection (RSICD) involves identifying differences between images captured within the same geographical area but at two distinct times [1]. As a prominent subject in remote sensing, RSICD has recently attracted considerable attention due to its extensive utility in various applications, such as disaster assessment [2], urban expansion [3], deforestation [4], and environmental monitoring [5], [6].

In recent years, transformer-based methods [7], [8], [9], [10], [11] have become one of the mainstream approaches to solving change detection problems due to their significant capability to model long-distance dependencies and achieve remarkable results on remote sensing images. However, transformer-based RSICD methods still have two issues: 1) current transformer-based change detection methods focus on long-distance modeling of unitemporal features, making it difficult to perceive the differences (e.g., edge texture and deep semantics) between bitemporal features. As a result, the change detection results of these methods exhibit significant missing, particularly when dealing with remote sensing images that contain multiple change regions or single large target changes and 2) the absence of an explicit spatial constraint imposed on the final fused bitemporal features leads the model to easily overlook feature representations of small targets during the classification, thus further reducing the detection performance.

The visual results of the above problems are shown in Fig. 1, which includes three complex change patterns: (I) multiple change regions, (II) single large target changes, and (III) small target changes. The prechange and postchange images are shown in Fig. 1(a) and (b), respectively. The detection results of two representative transformer-based change detection methods, BiT and ChangeFormer, are illustrated in Fig. 1(c) and (d). These two transformer-based methods struggle to accurately detect changes within multiple target regions or single large target regions due to insufficient exploration of the spatiotemporal relationships between bitemporal features. Additionally, the lack of explicit constraints on the final fused bitemporal features often leads these methods to overlook small target changes, especially when the content of the pair of remote sensing images is similar. These detection results further validate the challenges that existing transformer-based methods encounter when handling remote sensing images with complex change patterns. To analyze the detection results across the entire dataset, Fig. 1(f) gives a detailed statistical analysis of the distribution of misclassified pixels between predicted and actual change boundaries for BiT, Changformer, and our proposed SFBI-Net. The analysis reveals that the majority of false positive pixels (80%) and false negative pixels (70%) in the entire dataset are still located within a range of 3–7 pixels between the predicted and actual change boundaries, respectively. Hence, accurately distinguishing the difference between bitemporal features at identical network layers, while enhancing the feature representation of small targets, remains the principal challenge for current transformer-based change detection methods.

In this article, we propose a spatial focused bitemporal interactive network (SFBI-Net) for RSICD. SFBI-Net incorporates

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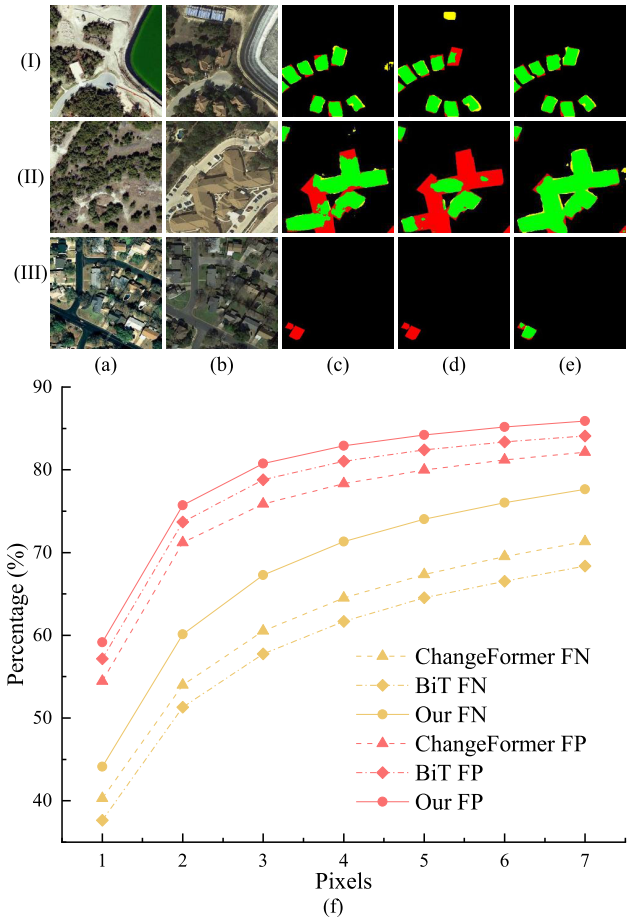


Fig. 1. Illustrating that the interaction between features in bitemporal images and the imposition of constraints on fused features are essential for current transformer-based methods in handling remote sensing images with complex change patterns. (a) Prechange image, (b) postchange image, (c) prediction results of the ChangeFormer, (d) prediction results of the BiT, (e) prediction results of our proposed SFBI-Net, and (f) statistical chart detailing the distribution of boundary pixel types. Distinct color coding is employed for visual clarity, with true positives represented by **green**, true negatives by **black**, false positives by **yellow**, and false negatives by **red**.

a bitemporal spatiotemporal interactive (BSI) module and a spatial focus diversity loss (SFD-Loss), which are designed to enhance the discriminative representation of bitemporal features and to improve the feature representation of small target areas, respectively. Specifically, the global feature interaction branches within the BSI module capture long-distance dependencies across temporal instances by calculating the global similarity of bitemporal features, while the convolutional branches supplement local detail information within a global receptive field. Moreover, an SFD-Loss is designed to enhance the feature representation of small targets by constraining the fused bitemporal features, thereby improving the model's detection performance on small targets. The proposed SFBI-Net addresses the problem of detection inaccuracies in existing transformer-based methods when handling remote sensing images with complex change patterns, such as multiple change regions, as shown in Fig. 1(e). It demonstrates superior performance compared to several state-of-the-art RSICD methods. The primary contributions of this work can be summarized as follows.

1) With a well-designed BSI module, global interactions and local information from bitemporal features are leveraged to capture differences between these features, enhancing change detection capabilities in remote sensing images, particularly in scenarios involving multiple change regions and single large target changes.

2) An SFD-Loss is proposed to explore how to enhance the representation of bitemporal features in the spatial dimension by constraining the fused feature to obtain the maximum feature representation, further enhancing the model's ability to detect small target areas.

3) We propose a novel method called SFBI-Net, which seamlessly integrates a well-designed BSI module and SFD-Loss. Experiments on challenging datasets, including LEVIR-CD [12], EGY-BCD [13], and CLCD [14], demonstrate that our method achieves superior performance in remote sensing images compared to several state-of-the-art methods.

The article is organized as follows. Section II presents an overview of related work on RSICD. Section III introduces details of the proposed methods, including the BSI module and the SFD-Loss. Section IV outlines a series of experiments conducted to evaluate the performance of the proposed method. Section V discusses the advances and future works of SFBI-Net, as well as the limitations of the current evaluation metric. Finally, Section VI concludes the article.

II. Related Work

In Section II, we explore the advancements in RSICD methods and transformer networks over the past several decades. Section II-A focuses on RSICD methods, while Section II-B introduces transformer networks.

A. Remote Sensing Image Change Detection

In the early stage of RSICD research, scholars primarily relied on hand-crafted feature modeling and analyzed differences among features to detect changes [15], [16], [17]. For instance, Wu et al. [15] proposed the SFA method, which transforms the original data into a new feature space, emphasizing the differences between changed pixels and enabling better separation of changed pixels. However, these methods exhibit limitations when applied to remote sensing images with complex scenes, as hand-crafted features are inadequate for representing intricate changes. Therefore, deep learning has emerged as the mainstream method for RSICD in recent years, due to its powerful feature representation ability and automatic feature learning approach.

In the past, the lack of sufficient labels to indicate changes between image pairs has promoted the development of unsupervised RSICD methods. Zhan et al. [18] proposed an unsupervised method based on the analysis of spatial–spectral change information. However, unsupervised deep learning methods often fail to accurately learn discriminative features due to the lack of an effective feedback mechanism. Consequently, image-level weakly supervised change detection methods have been developed [19], [20], [21]. Wu et al. [20] developed a change detection framework using a generative adversarial network for weakly supervised change detection.

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To mitigate the influence of noise in the pseudo-labels obtained through Class Activation Maps [22], Cao et al. [21] proposed a multiscale weakly supervised learning method for detecting changes in built-up areas. Furthermore, to address the limitation of image-level labels in providing accurate location information, Fang et al. [23] proposed a change detection network supervised by point-level labels to achieve more accurate change area localization and detection.

Compared to unsupervised methods, supervised deep learning methods demonstrate superior performance with the support of ground truth and have become a current primary research focus in RSICD. Liu et al. [24] introduced an approach that segments each image independently to obtain semantic labels and then utilizes the differences between these labels to identify the regions of ultimate change. Daudt et al. [25] proposed three fully convolutional Siamese neural networks (FC-EF, FC-Siam-Conc, and FC-Siam-Diff) trained end-to-end to generate pixel-level change maps from bitemporal images. Despite the significant progress achieved by these methods, they neglect the crucial role of region-of-interest features in change detection. As a result, the change maps generated by these methods often struggle to distinguish discriminative features, leading to inaccuracies in localizing and detecting changes.

Currently, the attention mechanism has been introduced into RSICD, enhancing the discriminative ability of the features extracted by the model. For example, Liu et al. [26] proposed the DTCDSCN model, which utilizes channels and spatial attention to improve the feature representation. Yang et al. [27] proposed SEDANet, which employs differential attention weight guidance for spatial and channel visual attention to reduce the influence of irrelevant change information. Shangguan et al. [28] introduced a graph convolutional neural network to capture long-range dependencies between detection targets and the global context and to enhance the representation of multiscale features through a multiscale spatial and channel attention fusion module. These methods significantly enhance the model's ability to perceive and detect changes. Additionally, various innovative approaches have been proposed to tackle the dual-input and complex scenarios characteristics of RSICD. Zhao et al. [29] proposed a new strategy that utilizes an exchange dual-encoder-decoder architecture to fuse bitemporal features and bitemporal semantic features for accurate change detection. Wen et al. [30] introduced GCD-DDPM, which utilizes the denoising diffusion probabilistic model to generate fine-grained change maps instead of classifying each pixel into changed or unchanged categories. Multilevel feature fusion is an effective approach to enhance a model's ability to perceive multi-scale changes. Ye et al. [31] proposed AFCF3D-Net, which leverages both the intrinsic fusion properties of 3-D convolutions and the AFCF module to mitigate the discrepancy between high-level semantic features and low-level spatial features, achieving more accurate detection results. To effectively leverage feature information from different network layers, Miao et al. [32] employed a Siamese U-Net network to extract features across multiple levels and used full-scale skip connections to ensure a more comprehensive feature representation. Furthermore, the remote sensing foundation model [33], [34] is a hot topic in current research. It acquires more generalizable knowledge through large-scale pretraining, facilitating better adaptation to various remote sensing scenarios and tasks. For example, Wang et al. [33] developed a foundational model for remote sensing through multitask pretraining on a large-scale remote sensing dataset.

B. Transformer Network

Transformer was introduced by Vaswani et al. [35] for machine translation, and has since become the state-of-the-art method in many natural language process tasks. The transformer effectively captures global contextual information and exhibits long-range dependencies, addressing the limitations of deep convolutional neural networks in capturing global information within features. Inspired by the remarkable success of the transformer, researchers have made efforts to apply this model to various computer vision tasks [36], [37], [38], [39], [40], [41]. For instance, Dosovitskiy et al. [36] demonstrated that a heavy reliance on CNNs is unnecessary. When directly applied to sequences of image patches, an original transformer architecture can achieve remarkable performance on image classification tasks. Carion et al. [38] proposed a novel object detection framework based on transformers, which utilizes bipartite matching loss for direct set prediction. In the field of semantic segmentation, Xie et al. [39] proposed a simple, clean yet powerful semantic segmentation method that incorporates a positional-encoding-free, hierarchical transformer encoder alongside a lightweight All-MLP decoder. The studies above have demonstrated the superior performance of the transformer architecture when utilized in image processing.

In RSICD, various transformer-based methods have also been widely proposed [7], [8], [11]. For instance, Chen et al. [7] integrated a transformer into a convolutional neural network to enhance the contextual modeling of high-level semantic information extracted by the convolutional neural network, thereby reinforcing semantic tokens. Subsequently, a transformer decoding structure is employed to decode these features and generate the final change maps. Pan et al. [11] proposed an M-Swin with a hierarchical window, fusing multiscale information to produce richer boundary details features. These methods benefit from the transformer's ability to capture long-distance dependencies, resulting in significant advancements in RSICD. Motivated by this success, we explore the spatiotemporal relationships between bitemporal images by leveraging the transformer's capability to capture long-distance dependencies, thereby effectively discerning differences between bitemporal features.

III. Methodology

In this section, we first elaborate on the proposed SFBI-Net, introduced in Section III-A, and then present the BSI module in Section III-B. Section III-C specifically illustrates the SFD-Loss and Section III-D details the total loss function.

A. Overview

The overall framework of our proposed SFBI-Net is illustrated in Fig. 2. SFBI-Net consists of two principal

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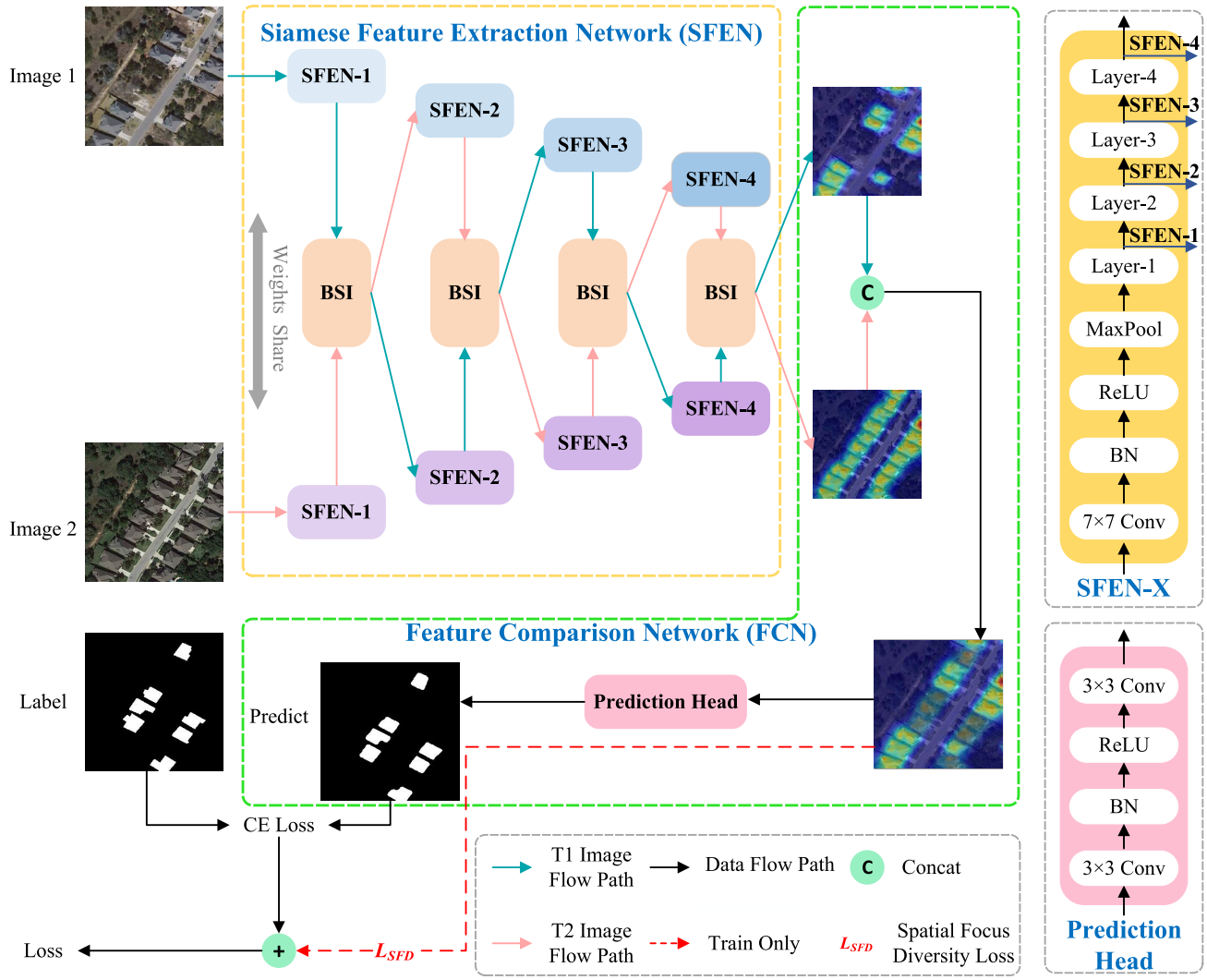


Fig. 2. Overall framework of our proposed SFBI-Net. The SFEN enhances the spatiotemporal relationships of bitemporal images for more discriminative feature representation through the BSI module. The BSI module introduces global interactive and local information of bitemporal features for more discriminative feature representation to better perceive changes. FCN can apply spatial constraints on the final fused bitemporal features through SFD-Loss, generating more accurate change maps. The seamless integration of SFEN and FCN enables our SFBI-Net to outperform other state-of-the-art methods.

components: the Siamese feature extraction network (SFEN) and the feature comparison network (FCN). In the first component, we use the BSI module to explore the spatiotemporal relationships between bitemporal images at the features within the same layer for discriminative feature representation. Four BSI modules separately model the spatiotemporal relationships of bitemporal features across four layers, ranging from shallow to deep feature layers, in the SFEN. These bitemporal features are extracted by a pretrained ResNet-18 [42], with SFEN-X ($X = 1, 2, 3, 4$) denoting the four distinct hierarchical layers, ranging from shallow to deep, in ResNet-18. Since the input bitemporal features produce cross-outputs after modeling the spatiotemporal relationships, we use blue and pink to designate the flow paths of Image 1 and Image 2, respectively, to distinguish the distinct bitemporal features during the extraction phase. In the second component, we fuse the discriminative bitemporal features modeled by the BSI modules and apply spatially focused constraints using SFD-Loss to enhance the representation of small targets, ultimately obtaining an accurate change map through the prediction head. The SFD-Loss is indicated by a red dashed line as a part of the overall loss function. The prediction head comprises two convolutional layers with 3×3 kernel size, batch normalization (BN), and ReLU activation function.

B. BSI Module

Recently, transformer-based methods [7], [8], [9], [43] have achieved considerable performance in change detection due to the transformer's effective capability to capture long-distance dependencies. However, the change detection results of these methods exhibit significant missing for bitemporal images with complex change patterns, such as multiple change regions or single large target changes, due to insufficient spatiotemporal correlation between the learned features from bitemporal images. Therefore, we introduce a BSI module to explore the spatiotemporal relationships among bitemporal features for discriminative feature representation, which better perceives changes. The BSI module comprises two batch

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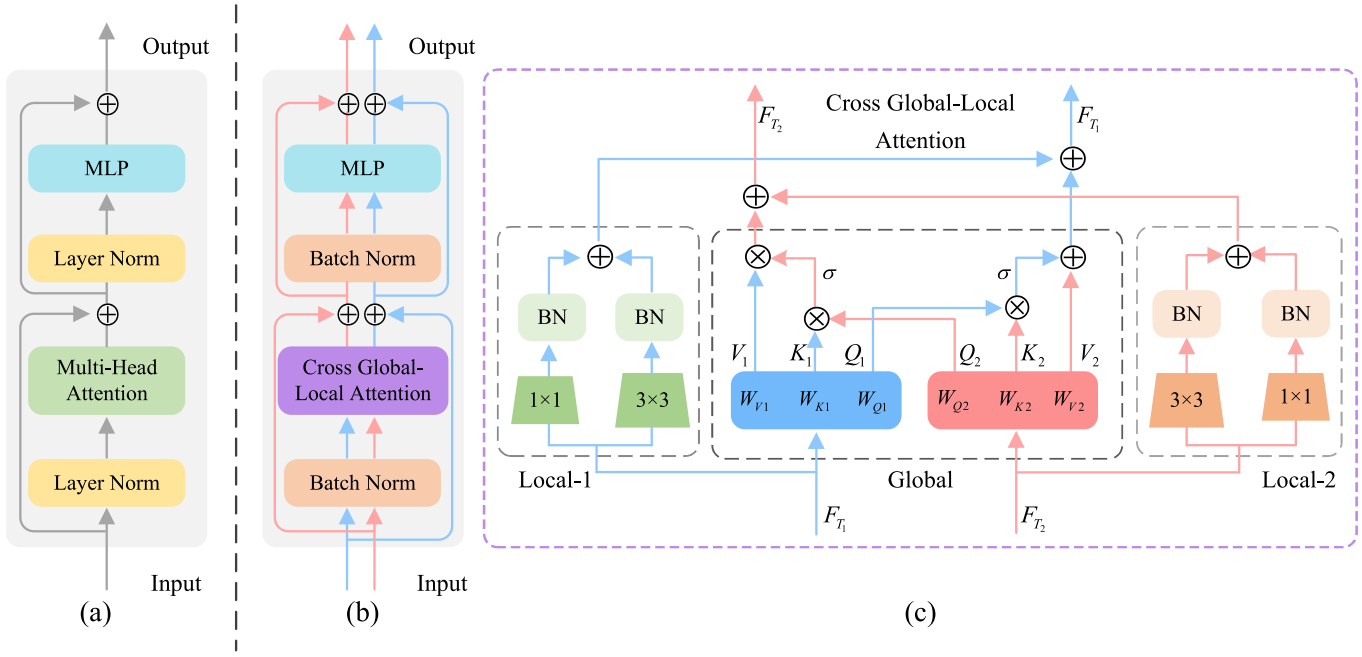


Fig. 3. (a) Standard transformer block, (b) BSI module, and (c) cross-global–local attention block. The global and local branches of the cross-global–local attention block are used for global feature interaction of bitemporal features and for enhancing local feature information within a global receptive field, respectively, resulting in discriminative bitemporal feature representations.

norm layers, a cross-global–local attention block, and an MLP layer, as shown in Fig. 3(b). The cross-global-local attention block is a pivotal component of the BSI module, facilitating global interactions among bitemporal features in a cross-query manner while enhancing their respective local detail information [see Fig. 3(c)]. Fig. 3 provides a visual comparison highlighting the distinct architectural differences between the BSI module and the standard transformer block. Compared to the standard transformer block shown in Fig. 3(a), the BSI module can handle dual-input images and capture the spatiotemporal relationships between them, enhancing feature discrimination and improving the model's ability to distinguish changes in remote sensing images.

1) Cross-Global–Local Attention: To capture the spatiotemporal relationships among bitemporal features and emphasize the detailed information that might be overlooked in a global context, we designed two branches in our approach: a global branch and a local branch. In the global branch, we use a cross-query approach to capture the similarity among bitemporal features. These similarities are then utilized to enhance the discriminative representation of the features, thereby more effectively distinguishing differences among bitemporal features. Meanwhile, the local branch employs two convolutional kernels of different sizes to extract detailed features within a local receptive field. By combining the outputs of these two branches, the CGLA block achieves a discriminative feature representation that captures the global interaction between bitemporal features and enhances the detailed information that is easily lost within a global receptive field. The specific process is described as follows.

a) Global branch: Given bitemporal features $T_1 \in \mathbb{R}^{L\times C}$ and $T_2 \in \mathbb{R}^{L\times C}$ as input, where $L = W \times H$ with H, W, and C representing the height, width, and number of channels, respectively. Bitemporal features T_1 and T_2 are individually processed through learning parameters W_{v1}, W_{q1}, W_{k1}, and W_{v2}, W_{q2}, W_{k2} to obtain their respective query (Q), key (K), and value (V) representations. Subsequently, a cross-query approach is applied to model the bitemporal features and enhance their respective representations. The formulation for this process is as follows:

$$\mathrm{CGLA}_{\mathrm{global}-1}(T_1, T_2) = \sigma_c\left(\frac{Q_{T_1}K_{T_2}^{\mathrm{T}}}{\sqrt{C'}}\right)V_2 \tag{1}$$

$$\mathrm{CGLA}_{\mathrm{global}-2}(T_2, T_1) = \sigma_c\left(\frac{Q_{T_2}K_{T_1}^{\mathrm{T}}}{\sqrt{C'}}\right)V_1 \tag{2}$$

where $\mathrm{CGLA}_{\mathrm{global}-1}$ and $\mathrm{CGLA}_{\mathrm{global}-2}$, respectively, represent the T_1 and T_2 feature representations after being enhanced with global interaction. $\sigma_c(\cdot)$ denotes the softmax operation applied to the channel dimension and C' is the channel dimension of Q, K, and V.

b) Local branch: This branch supplements the local information of bitemporal features, preventing the model from overlooking fine details in the interaction of bitemporal features within a global receptive field. We use 1×1 and 3×3 convolutional blocks to extract local features from different receptive fields, respectively. Subsequently, we integrated these two local features to obtain the final detailed feature representation as follows:

$$\mathrm{CGLA}_{\mathrm{local}-1}(T_1) = \mathrm{BN}\left(W_{1\times1}^{1}T_1\right) + \mathrm{BN}\left(W_{3\times3}^{2}T_1\right) \tag{3}$$

$$\mathrm{CGLA}_{\mathrm{local}-2}(T_2) = \mathrm{BN}\left(W_{1\times1}^{1}T_2\right) + \mathrm{BN}\left(W_{3\times3}^{2}T_2\right) \tag{4}$$

where $\mathrm{CGLA}_{\mathrm{local}-1}$ and $\mathrm{CGLA}_{\mathrm{local}-2}$, respectively, represent the T_1 and T_2 local feature representation. $W_{1\times1}^{1}$ and $W_{3\times3}^{2}$

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Algorithm 1 CGLA Block Algorithm

Input: Unchanged feature: T_1; Changed feature: T_2;
Output: Enhanced features: T_1, T_2;
Initialize the parameters of CGLA block θ randomly;
for $t = 1, \ldots, 200$ *epochs* **do**
  1. Global Branch
  Calculate the spatiotemporal correlation between feature maps T_1 and T_2, and model the feature context according to equation. 1 and equation. 2
  2. Local Branch
  Enhance local features according to equation. 3 and equation. 4
  3. Bitemporal feature discrimination representation
  Integrate global and local branches of bitemporal features to enhance discrimination representation according to equation. 5 and equation. 6
  Update θ

represent the parameters of two convolutional kernels with different sizes.

c) Bitemporal feature discrimination representation: We separately fused the bitemporal features from these two different receptive fields to achieve more discriminative feature representations. The fused process is as follows:

$$T_1 = \mathrm{CGLA}_{\mathrm{local}-1}(T_1) + \mathrm{CGLA}_{\mathrm{global}-1}(T_1, T_2) \tag{5}$$

$$T_2 = \mathrm{CGLA}_{\mathrm{local}-2}(T_2) + \mathrm{CGLA}_{\mathrm{global}-2}(T_2, T_1) \tag{6}$$

where T_1 and T_2 represent the bitemporal feature representations resulting from the fusion of local and global information. This discriminative feature representation ensures that our model can learn information biases from global and local perspectives, providing a better perception of differences in bitemporal features. Based on the above description, the implementation of the CGLA block is summarized in Algorithm 1.

C. Spatial Focus Diversity Loss

Current mainstream change detection approaches [7], [9], [43] commonly employ feature fusion strategies to enhance the bitemporal feature representations at the decoding layer, followed by pixel-wise category prediction by a classification head. While the fused features have integrated contextual information, directly classifying these features has limitations due to the lack of constraints on the fusion features. Specifically, the model tends to focus on discriminative change areas during classification while overlooking small target areas, which poses a challenge for accurate detection. More recently, imposing constraints on features [44], [45] has been shown to be useful in obtaining more representative features. However, effectively imposing constraints on features to enhance the representation of small target areas is still challenging. Hence, in this article, an SFD-Loss is proposed to constrain the fused features for more representative features and enhance the feature representation of small targets.

First, considering the differences in the spatial distribution of features across different channels, directly imposing constraints on the features may easily overlook these disparities. We need to capture the inherent global interdependencies among the feature channels to distinguish channels with differences, as described below.

Given a fused feature $A \in \mathbb{R}^{C\times H\times W}$, where C, H, and W represent the number of channels, height, and width, respectively, we reshape A to $B \in \mathbb{R}^{C\times N}$ to calculate the global channel information correlation weight matrix, where $N = H \times W$. Subsequently, the crucial channels within the original feature map associated with the global context are enhanced by performing matrix multiplication of the weight matrix with A. This operation produces the final feature map $\hat{F} \in \mathbb{R}^{C\times H\times W}$

$$\hat{F} = \left(\sigma_c\left(BB^{\mathrm{T}}\right)\right)A \tag{7}$$

where $\sigma_c(\cdot)$ is the softmax operation operated on the channel dimension, and the weight matrix measures the relationships among global feature channels, having a shape of $C \times C$.

After that, to maximize feature representation in the spatial dimension and indirectly enhance feature diversity, we divide the feature map $\hat{F}$ into k semantic categories. The channels in each semantic category serve as the smallest constrained unit, and the ith semantic category is defined as follows:

$$f_i = \sum_{j=0}^{\delta-1} \hat{F}_{i\times\delta+j} \tag{8}$$

where δ indicates the number of feature channels encompassed within each category. We initially perform spatial feature normalization to obtain spatial weights representation for each semantic category, followed by max-pooling operations across channel dimensions within each semantic category to extract the most salient spatial representation, formalized as follows:

$$D = \mathrm{maxpool}_{c=0}^{k-1}\left(\sigma_s\left(\hat{F}\right)\right) \tag{9}$$

where $D \in \mathbb{R}^{k\times H\times W}$ represents the spatial representation of k semantic categories and $\sigma_s(\cdot)$ denotes the softmax operation applied on the spatial dimension. This operation produces feature weights ranging from 0 to 1 to emphasize the significance of spatial features. We apply max-pooling operations in the spatial dimension to suppress interference from background regions. Finally, we calculate the mean of the discriminative features across various semantic categories as the SFD-Loss, defined as follows:

$$\Upsilon = -\frac{1}{k}\sum_{i=0}^{k-1} \mathrm{maxpool}(D) \tag{10}$$

where Υ represents the final feature constraint, which is a gradient component that helps to enlarge the inter-class distance of all targets. Furthermore, the increase in the minimum inter-class distance on the feature map demonstrates the effectiveness of our constraints in enhancing discriminative features and achieving the objective of maximizing feature values within the spatial dimension. As shown in Fig. 4, the intuitive feature constraint results are apparent. Compared to the feature map before applying SFD-Loss [see Fig. 4(a)],

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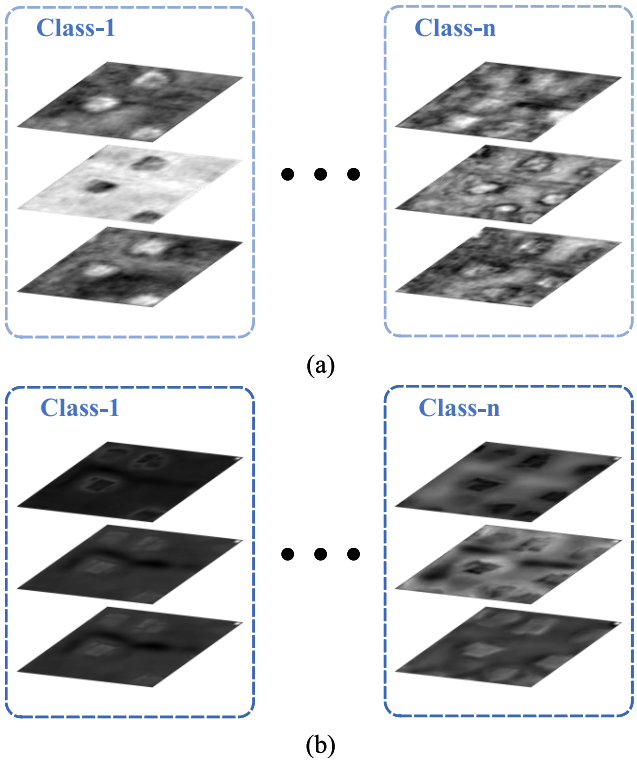


Fig. 4. (a) Comparison of feature maps before applying SFD-Loss and (b) after applying SFD-Loss, where different semantic categories emphasize distinct features within the channel of interest. The SFD-Loss effectively directs the model to focus on different discriminative regions across feature channels for distinct categories. This diversity in feature representation enhances the model's detection performance, particularly in small target regions.

the feature map generated by the model after SFD-Loss [see Fig. 4(b)] exhibits larger feature values in the spatial dimension, and different semantic categories emphasize distinct features within the channel of interest. In other words, after applying SFD-Loss, the model enhances the feature representation of previously overlooked regions, such as small targets, and indirectly increases the diversity of the features.

D. Total Loss Function

We represent the predicted change map as P and the ground truth as Y. We trained the proposed model by optimizing the cross-entropy loss L_{ce} and the SFD-Loss Υ. The total loss function is defined as follows:

$$L_{\text{ce}}(Y, P) = -\frac{1}{N}\sum_{i}^{N}\left[Y_i \log_2 P_i + (1 - Y_i)\log_2(1 - P_i)\right] \tag{11}$$

$$L_{\text{total}} = L_{\text{ce}} + \lambda\Upsilon \tag{12}$$

where N represents the total number of pixels in the final change map, P_i represents the probability value predicted by the model for pixel i belongs to the change class, and Y_i represents the ground true value of pixel i. λ is the weighting factor applied to regulate the influence of the SFD-Loss.

IV. Experiments

In this section, we evaluate the effectiveness of the proposed SFBI-Net on three RSICD datasets: the LEVIR-CD dataset, the CLCD dataset, and the EGY-BCD dataset. The datasets and implementation details are introduced in Sections IV-A and IV-B, respectively. Subsequently, we describe the state-of-the-art methods in Section IV-C and provide a performance comparison with these methods in Section IV-D. To validate the effectiveness of each component of the proposed SFBI-Net, a comprehensive ablation analysis is provided in Section IV-E.

A. Datasets

1) LEVIR-CD: is a publicly available large-scale building change detection dataset consisting of 445 pairs in the training set, 64 pairs in the validation set, and 128 pairs in the test set. Each pair consists of bitemporal images with a size of 1024 × 1024. To ensure a fair comparison with other methods and account for GPU memory limitations, each image in this dataset is partitioned into 16 nonoverlapping image patches, each sized at 256 × 256. As a result, 7120 pairs of image patches are used for training, 1024 pairs for validation, and 2048 pairs for testing.

2) EGY-BCD: is developed to detect building changes from satellite imagery at a resolution of 0.25 m/pixel. It comprises 6091 pairs of images, each sized at 256 × 256, and is randomly divided into three parts: a training set (70%), a validation set (20%), and a test set (10%).

3) CLCD: is a publicly available cropland change detection dataset that comprises 600 pairs of cropland change samples. This study randomly divides images into three parts: 360 for training, 120 for validation, and 120 for testing. We also employed a nonoverlapping approach to crop the images to 256 × 256. As a result, 480, 480, and 1280 pairs of image patches are utilized for validation, testing, and training, respectively. Unlike the previously mentioned two datasets that primarily address building changes, CLCD includes various changes, such as those in buildings, roads, lakes, and bare soil lands.

B. Experiment Implementation Details

The proposed SFBI-Net is implemented in the PyTorch framework and trained on an NVIDIA GeForce RTX 3090 GPU. We utilized stochastic gradient descent with momentum to optimize the model, setting the initial learning rate to 0.01, the momentum to 0.99, the weight decay to 1e-5, and the linear decay to 0 over 200 epochs. The batch size is set to 4, and the input size for the three datasets is 256 × 256 × 3. We applied the same data augmentation techniques, including flipping, cropping, and random adjustment of brightness and contrast. The weight value of the SFD-Loss λ is set to 0.01.

Five common metrics are employed to evaluate the performance of the proposed method: precision, recall, $F1$-score, overall accuracy (OA), and intersection over union (IoU). Specifically, higher precision indicates more accurate detection of change pixels, while higher recall signifies the network's ability to detect more change. OA represents the ratio of pixels accurately categorized across all categories. IoU represents the degree of overlap between the region predicted by the model and the annotated region in the label. $F1$ is the harmonic mean

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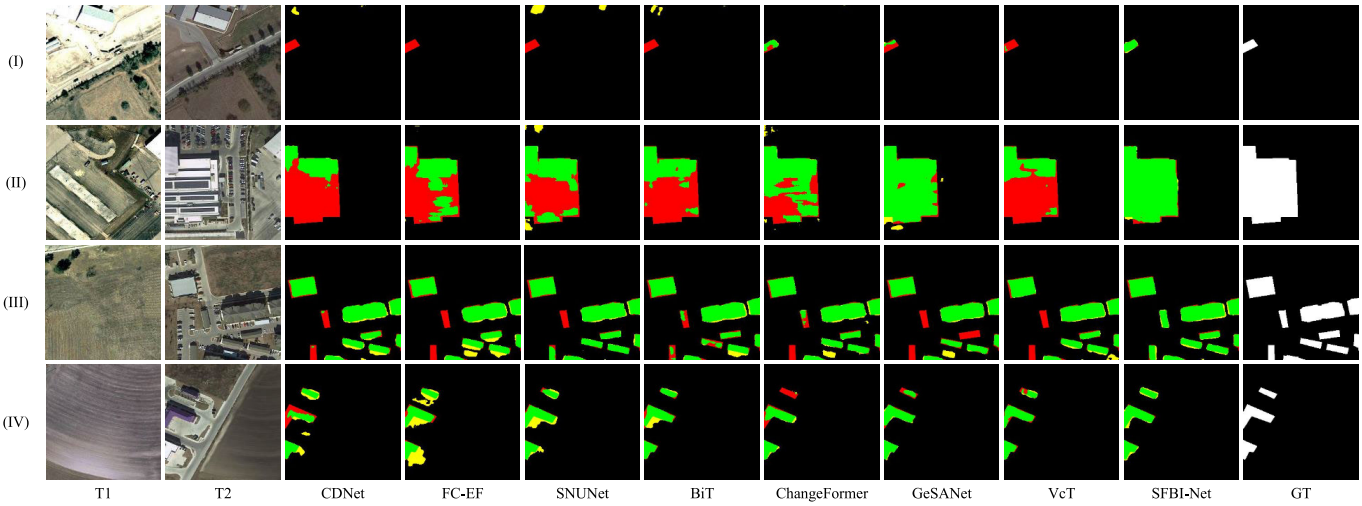


Fig. 5. Visual results of our proposed SFBI-Net and other methods on the LEVIR-CD test set. Various colors are utilized for clarity, i.e., **green** represents true positives, **black** represents true negatives, **yellow** represents false positives, and **red** represents false negatives. This color scheme remains consistent in subsequent figures.

of precision and recall and is our main evaluation metric. The metrics above are defined as follows:

$$\text{Precision} = \text{TP}/(\text{TP} + \text{FP}) \tag{13}$$

$$\text{Recall} = \text{TP}/(\text{TP} + \text{FN}) \tag{14}$$

$$\text{F1} = 2\text{TP}/(2\text{TP} + \text{FP} + \text{FN}) \tag{15}$$

$$\text{IoU} = \text{TP}/(\text{TP} + \text{FN} + \text{FP}) \tag{16}$$

$$\text{OA} = (\text{TP} + \text{TN})/(\text{TP} + \text{TN} + \text{FN} + \text{FP}) \tag{17}$$

where TP denotes the number of accurately identified changed pixels, while FP represents the number of unchanged pixels mistakenly detected as changed. TN signifies the number of unchanged pixels detected correctly, and FN denotes the number of changed pixels mistakenly identified as unchanged.

C. Comparison With State-of-the-Art Methods

To validate the effectiveness of our method, we conduct a comparison with nine state-of-the-art methods, including CDNet [46], FC-EF [25], FC-Siam-Diff [25], FC-Siam-Conc [25], SNUNet [47], BiT [7], ChangeFormer [8], GeSANet [48], and VcT [43]. Among them, CDNet is a typical change detection network based on the FCN architecture. FC-EF, FC-Siam-Diff, and FC-Siam-Conc are three networks also based on the FCN architecture, introduced in the same research paper, and differ in the fusion stages between bitemporal images. The SNUNet is a densely connected U-shaped Siamese network that incorporates a channel attention module to refine missing detailed features in semantic information. BiT is a hybrid model that combines convolutional neural networks and transformer architectures, enriching the high-level semantic tokens extracted by the convolutional neural network in both temporal and spatial domains through the transformer architecture, enhancing the availability of semantic information. ChangeFormer is a hierarchical transformer Siamese network with a lightweight decoder and represents a shift in change detection methods from convolutional neural networks toward transformer-based architectures. GeSANet is a Siamese change detection network that integrates geospatial position matching mechanisms and geospatial content matching mechanisms to mitigate the interference caused by misregistration and pseudo-change information resulting from various manual and natural factors. VcT is a Siamese change network that enhances the change detection process by exploiting the common background feature representation between two images, enabling precise differentiation between foreground and background changes.

D. Performance Comparison

1) Analysis of Results on the LEVIR-CD Dataset: Table I presents the quantitative comparison results with other methods on the LEVIR-CD dataset. The table demonstrates that the proposed SFBI-Net achieves outstanding results in four metrics. Specifically, the SFBI-Net achieves recall, $F1$, IoU, and OA values of 89.52%, 90.80%, 83.15%, and 99.07%, respectively. While VcT achieves the highest precision, it exhibits limitations in learning discriminative features, resulting in unsatisfactory performance in other metrics such as recall and OA. Specifically, compared to SFBI-Net, VcT's recall and OA are 1.87% and 0.76% lower, respectively.

We present the representative visual results of the proposed SFBI-Net and other methods on the LEVIR-CD dataset in Fig. 5. Notably, the comparative methods FC-EF, FC-Siam-Conc, and FC-Siam-Diff, introduced in the same research paper, differ only in their feature fusion approach. For a more intuitive comparison, we selected the best visualization results from these three methods to serve as a reference, as is the case in subsequent comparative figures. The four rows of comparative results in the figure represent the outstanding performance of our proposed method across four distinct scenarios. In row (I), SFBI-Net can accurately detect small target change areas between two content-similar remote sensing images, which is challenging for other methods. When large-scale area changes occur in a pair of images, SFBI-Net can detect the entire change area, closely aligning with the ground truth. In contrast,

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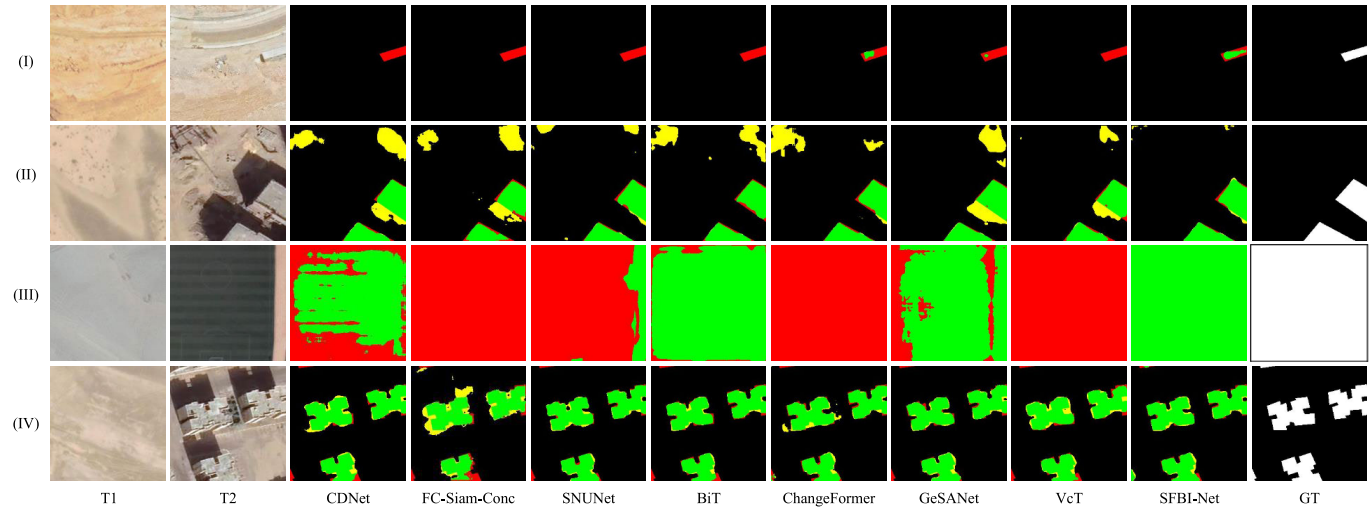


Fig. 6. Visual comparison of the results of our proposed SFBI-Net and other methods on the EGY-BCD test set.

TABLE I

QUANTITATIVE RESULTS ON THE LEVIR-CD DATASET ARE PRESENTED BELOW. (THE BEST PERFORMANCE IS HIGHLIGHTED IN **BOLD**, THE SECOND-BEST IS HIGHLIGHTED IN *ITALICIZED*, AND THE THIRD-BEST IS HIGHLIGHTED IN UNDERLINED, AS IS THE CASE IN SUBSEQUENT METHOD COMPARISON TABLES)

Method	Precision(%)	Recall(%)	F1(%)	IoU(%)	OA(%)
CDNet	90.50	85.95	88.17	78.84	98.83
FC-EF	88.80	81.36	84.92	73.79	98.53
FC-Siam-Conc	90.46	84.91	87.60	77.93	98.78
FC-Siam-Diff	91.89	81.13	86.18	75.71	98.67
SNUNet	89.18	87.17	88.16	78.83	98.82
BiT	89.24	*89.37*	89.31	80.68	98.92
ChangeFormer	92.05	88.80	*90.40*	82.48	*99.04*
GeSANet	92.04	88.15	90.05	*82.56*	99.01
VcT	**92.57**	87.65	90.04	*81.89*	99.01
SFBI-Net(ours)	*92.11*	**89.52**	**90.80**	**83.15**	**99.07**

TABLE II

QUANTITATIVE RESULT ON THE EGY-BCD DATASET

Method	Precision(%)	Recall(%)	F1(%)	IoU(%)	OA(%)
CDNet	81.62	72.47	76.77	62.30	91.53
FC-EF	68.68	72.37	70.48	54.41	88.29
FC-Siam-Conc	68.01	72.84	70.34	54.25	88.14
FC-Siam-Diff	70.19	73.41	71.76	55.96	88.84
SNUNet	*83.33*	*78.14*	*80.65*	*67.57*	*92.76*
BiT	81.79	74.51	77.98	63.91	91.88
ChangeFormer	82.09	75.37	78.59	64.72	92.07
GeSANet	82.63	75.41	78.85	65.89	92.19
VcT	75.95	71.86	73.85	58.54	90.17
SFBI-Net(ours)	**84.08**	**78.99**	**81.45**	**68.71**	**93.05**

other methods such as ChangeFormer and GeSANet can detect most of the changed areas but struggle to identify the entire changed area, as demonstrated in row (II). In row (III), SFBI-Net demonstrates exceptional proficiency in accurately predicting changed regions, even in multiple changed areas. In contrast, other methods such as CDNet and VcT exhibit varying degrees of missed detections in small-scale areas. While most methods, such as FC-EF, SNUNet, and BiT, tend to produce false detections due to background influence, SFBI-Net is less affected by these influences and demonstrates robust boundary learning ability, as shown in row (IV).

2) Analysis of Results on the EGY-BCD Dataset: The results of the quantitative comparisons on the EGY-BCD dataset are displayed in Table II. Notably, SFBI-Net exhibits optimal performance across multiple metrics, achieving a precision of 84.08%, a recall of 78.99%, an $F1$ of 81.45%, an IoU of 68.71%, and an OA of 93.05%. It outperforms the optimal SNUNet with improvements of 0.75% in precision, 0.85% in recall, 0.5% in $F1$, 1.14% in IoU, and 0.29% in OA.

A visual comparison of four pairs of qualitative results obtained from the EGY-BCD dataset test sample is presented in Fig. 6. In row (I), SFBI-Net demonstrates a remarkable ability to detect small targets, whereas other methods, including CDNet and FC-Siam-Conc, may overlook these change objects. Although ChangeFormer and GeSANet can detect changed objects, their detection results are not as pronounced as those of SFBI-Net. In row (II), SFBI-Net is less affected by non-building backgrounds and can accurately identify changed areas, as evidenced by the lack of false detection areas (highlighted in yellow) compared to other methods such as GeSANet and VcT. While the EGY-BCD dataset primarily focuses on detecting changes in buildings, it also includes a few other types of changes, such as soccer fields. As shown in row (III), ChangeFormer, FC-Siam-Conc, and VcT struggled to detect changes. CDNet, SNUNet, and GeSANet were able to identify small portions of the changed area. While BiT achieves relatively satisfactory detection results, SFBI-Net can detect the entire changed area, closely aligning with the ground truth labels. In detecting changes across multiple regions, SFBI-Net can accurately detect changed areas and their edge contours better than other methods such as VcT and GeSANet, as shown in row (IV).

3) Analysis of Results on the CLCD Dataset: The results of the quantitative analysis of all compared methods on the CLCD dataset are presented in Table III. The proposed SFBI-Net has achieved optimal results except for the precision metric. The $F1$ and IoU metrics are 2.6% and 2.93% higher than the second-place SNUNet. BiT employs a transformer

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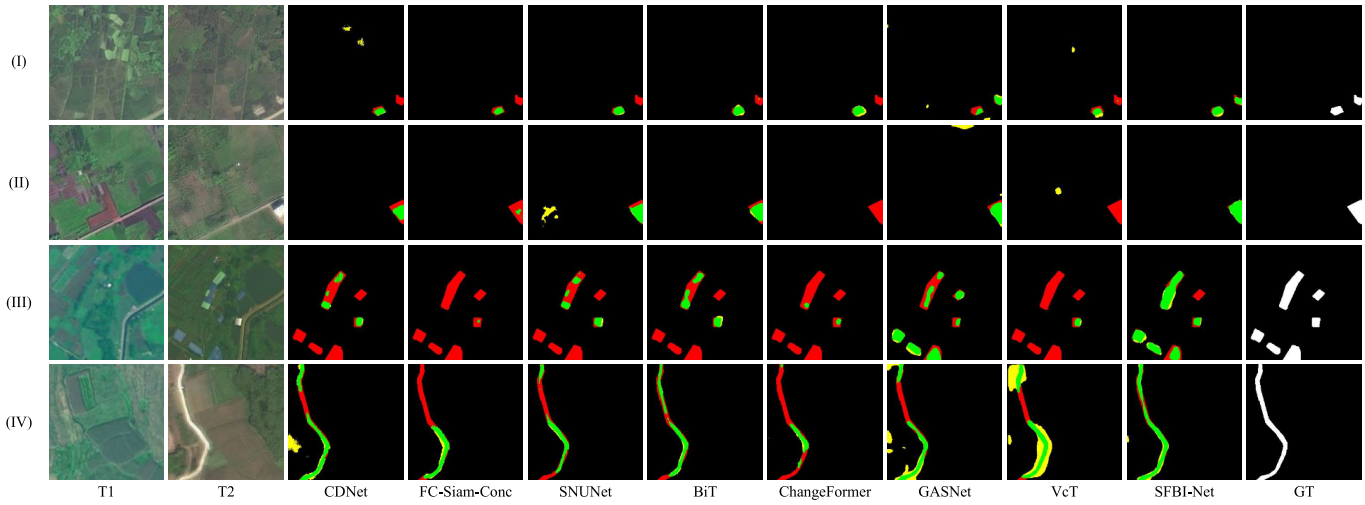


Fig. 7. Visual comparison of the results of our proposed SFBI-Net and other methods on the CLCD test set.

TABLE III
QUANTITATIVE RESULT ON THE CLCD DATASET

Method	Precision(%)	Recall(%)	F1(%)	IoU(%)	OA(%)
CDNet	65.95	58.07	61.76	45.19	94.66
FC-EF	66.41	55.21	60.18	40.07	89.37
FC-Siam-Conc	65.27	57.61	61.20	42.60	93.69
FC-Siam-Diff	64.24	59.17	61.60	44.32	93.37
SNUNet	63.91	*66.59*	*65.22*	*48.39*	94.72
BiT	**69.38**	58.08	63.03	46.23	94.96
ChangeFormer	65.04	63.85	64.44	41.87	94.42
GeSANet	*68.00*	60.21	63.87	47.12	*95.03*
VcT	52.68	40.01	45.52	29.47	92.86
SFBI-Net(ours)	66.31	**69.40**	**67.82**	**51.31**	**95.10**

TABLE IV
ABLATION STUDY OF THE FUNDAMENTAL COMPONENTS WITHIN OUR PROPOSED SFBI-NET ON THE LEVIR-CD DATASET

Index	Baseline	BSI	SFD-Loss	F1(%)	IoU(%)	OA(%)
1	✓	✗	✗	88.13	80.33	92.82
2	✓	✓	✗	90.52	82.69	99.06
3	✓	✗	✓	89.27	80.62	98.93
4	✓	✓	✓	**90.80**	**83.15**	**99.07**

architecture to enrich semantic information, resulting in its first-place position in the precision metric. However, the lack of semantic comparisons between bitemporal images hinders its recall metric, resulting in a score that is 11.32% lower compared to SFBI-Net. The performance of VcT is unsatisfactory, as it struggles to detect changes effectively in remote sensing images. Despite this, VcT achieves a relatively high OA metric compared to its $F1$ metric, primarily due to the low proportion of changes within the dataset. It is worth noting that change detection on the CLCD dataset presents significant challenges primarily due to two factors: a wide range of image resolutions (0.5–2 m) and diverse types of change categories, including buildings, land, and roads.

Fig. 7 shows the representative visualization results from existing methods and our SFBI-Net on the CLCD dataset. Rows (I) and (II) demonstrate that most models, such as ChangeFormer and VcT, exhibit instances of missed detections (highlighted in red), while SFBI-Net shows superior performance in change detection with no missed detections, particularly for small target areas. The low resolution of remote sensing images makes detecting changes in multiple small targets within agricultural fields challenging. Nevertheless, SFBI-Net can detect the majority of these changes, as shown in row (III). In row (IV), the performance of different methods in detecting road-type changes was compared. While methods such as CDNet and FC-Siam-Conc can detect changes in roads, none exhibit the completeness of detection by SFBI-Net.

E. Ablation Study

To verify the effect of each component in SFBI-Net, we conducted corresponding ablation experiments on the LEVIR-CD dataset. Specifically, we added the BSI module and SFD-Loss to the baseline, indicated by ticks and crosses. The baseline is a fundamental change detection network built upon a pretrained ResNet-18 Siamese architecture. The results of the ablation experiments are presented in Table IV, with the optimal results emphasized in bold.

We gradually added these components to assess their impact on the final change map, from indices 1 to 4. Compared to the baseline, which added the BSI module at index 2, there were varying degrees of improvement, specifically, $F1$ (2.39%), IoU (2.36%), and OA (6.24%). This finding confirms the efficacy of the BSI module within the proposed SFBI-Net. The primary enhancement arises from the model's ability to acquire more discriminative features, as evidenced by Fig. 8, where the feature attention regions gradually align with the ground truth. The baseline, augmented with the SFD-Loss at index 3, exhibited increments of 1.14%, 0.26%, and 6.11% in $F1$, IoU, and OA metrics, respectively, compared with index 1. Nevertheless, this only evaluates the efficacy of our proposed SFD-Loss without providing a qualitative evaluation of the enhancement conferred by our approach to the final results. Therefore, Fig. 9 shows the visual results of the algorithm for indexes 1 and 3. By comparing the d-column (index

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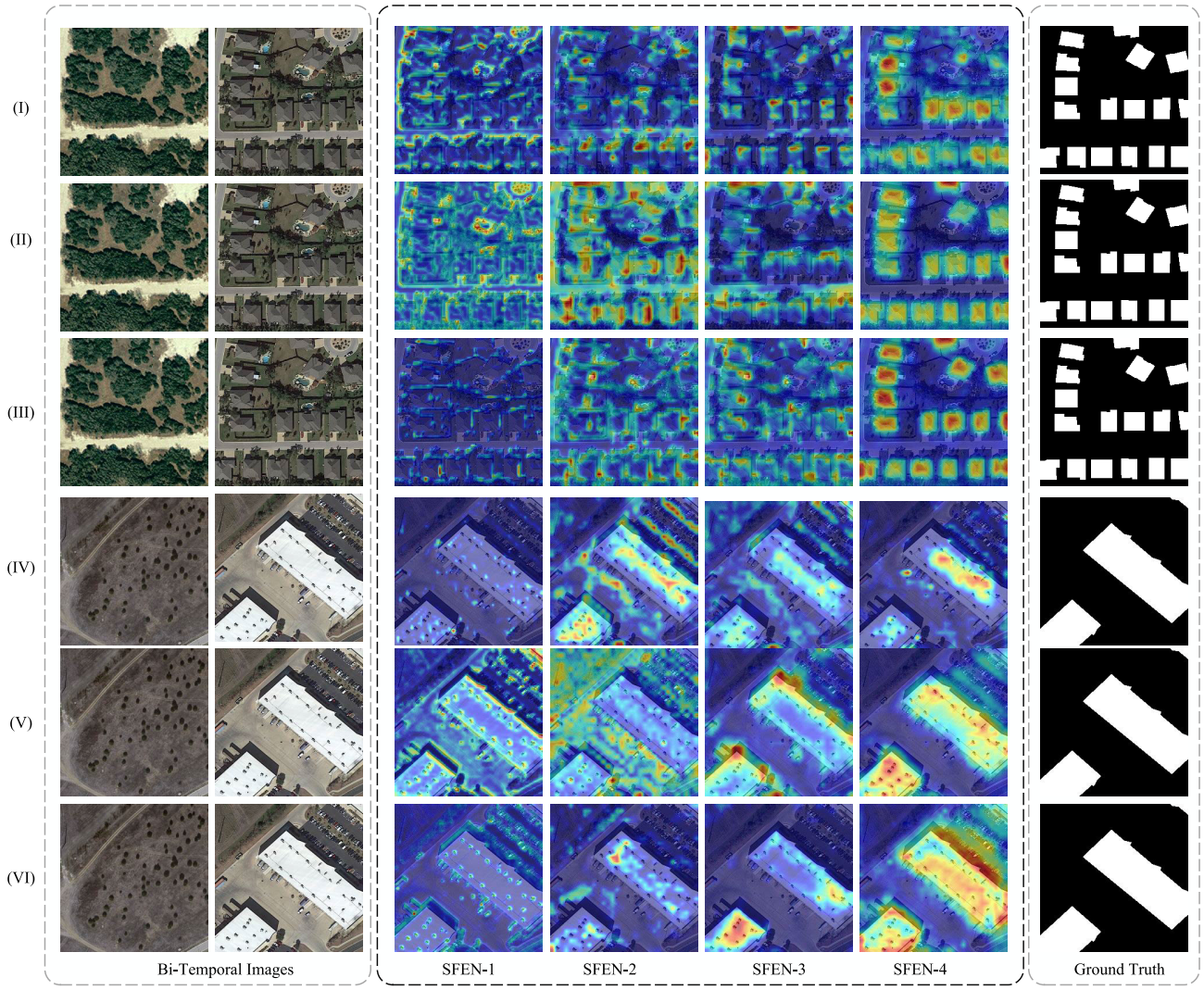


Fig. 8. Feature visualization examples, including multiple change regions and single large target changes, are presented on the LEVIR-CD test set, where blue and red denote lower and higher attention values, respectively. The four SFEN stages (SFEN-1, SFEN-2, SFEN-3, SFEN-4) correspond to distinct hierarchical levels of ResNet-18. Rows I and IV represent two baseline pairs without the BSI module integrated. In contrast, rows II and V depict two baseline pairs with the BSI module incorporated at different stages. Finally, rows III and VI showcase two baseline pairs with the BSI module added at all stages.

1) and e-column (index 3) in the images, it is evident that our proposed SFD-Loss significantly enhances the model's detection performance on small target regions. Considering the limited number of small target change regions in the dataset, this indirectly confirms the relatively limited effectiveness of this approach compared to the BSI module. Ultimately, our proposed SFBI-Net achieves optimal performance when all modules are integrated at index 4. The experimental results demonstrate that each component plays a crucial role in improving the detection performance of SFBI-Net.

1) Analysis of the BSI Module Across Different Stages of Feature Extraction: To investigate the influence of integrating the BSI module at distinct feature extraction stages, we systematically tested its integration into the four SFEN stages (SFEN-1, SFEN-2, SFEN-3, SFEN-4). Each stage corresponds to a distinct hierarchical level of ResNet-18, encompassing a range of shallow to deep feature representations. For instance, SFEN-1 corresponds to layer-1 of the ResNet-18, as depicted in Fig. 2. As shown in Table V, integrating the BSI module

TABLE V

ABLATION EXPERIMENT OF OUR PROPOSED BSI MODEL METHOD ON THE LEVIR-CD DATASET

Index	SFEN-1	SFEN-2	SFEN-3	SFEN-4	F1(%)	IoU(%)	OA(%)
1	✗	✗	✗	✗	88.13	80.33	92.82
2	✓	✗	✗	✗	90.13	82.04	99.00
3	✗	✓	✗	✗	90.29	82.30	99.03
4	✗	✗	✓	✗	90.49	82.63	99.05
5	✗	✗	✗	✓	90.33	82.36	99.04
6	✓	✓	✓	✓	**90.52**	**82.69**	**99.06**

at any stage of the SFEN feature extraction process resulted in varying degrees of improvement. Optimal detection performance was observed when the BSI module was incorporated at four stages. This finding is consistent with our hypothesis that the BSI module can capture discriminative features throughout the entire feature learning process (from edge textures to deep semantics) by leveraging its bitemporal feature global interaction. Further analysis reveals that integrating the BSI module at a deeper stage has a more significant effect due to deep features containing more semantic information, as shown

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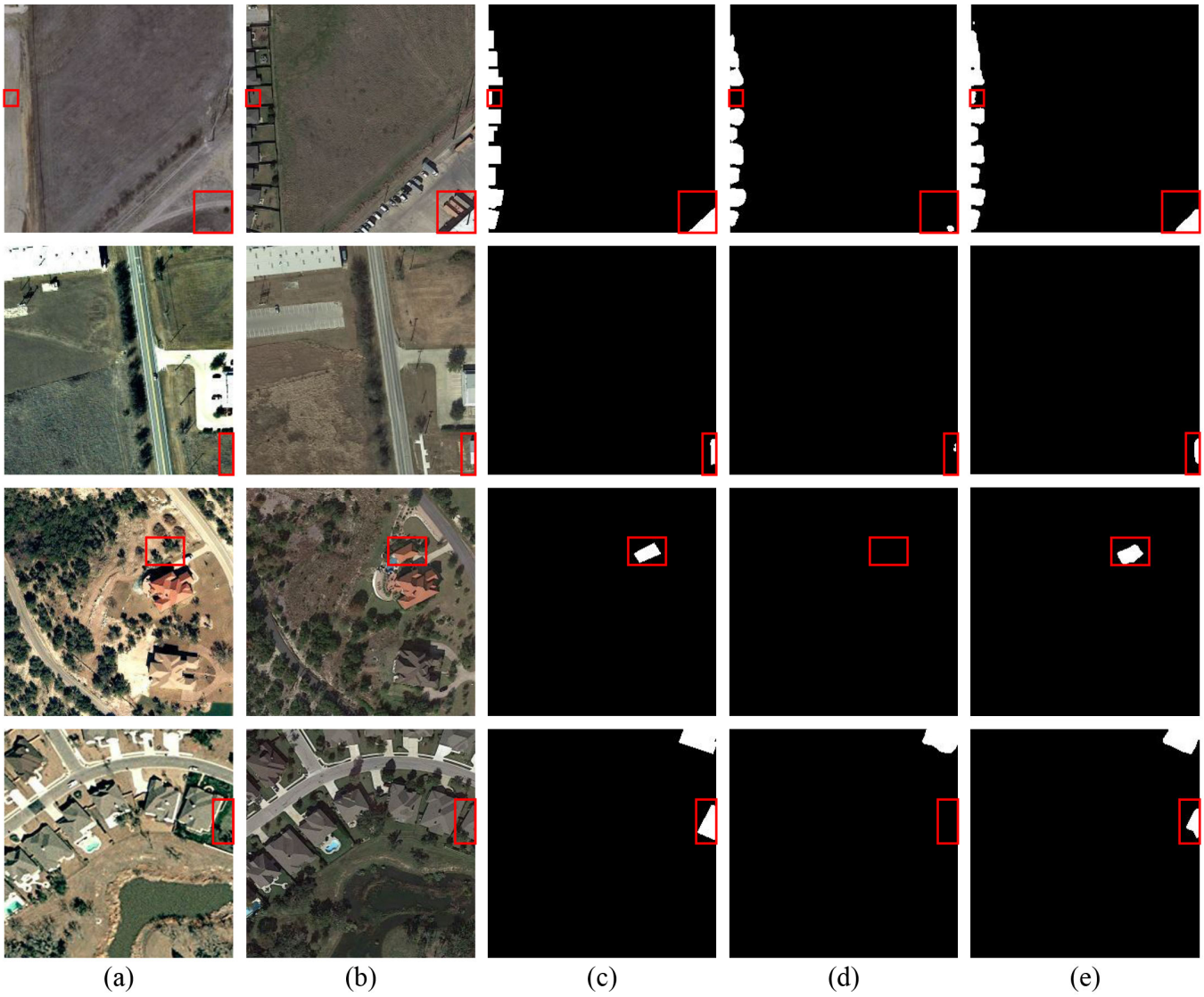


Fig. 9. Examples of SFD-Loss ablation experiments on the LEVIR-CD test dataset. (a) Prechange image. (b) Postchange image. (c) Ground truth. (d) Change map without SFD-Loss. (e) Change map with SFD-Loss. The red boxes indicate small objects that are challenging for the model to detect.

in indexes 4 and 5. Despite index 5 encompassing richer semantic information than index 4, the model's detection metrics are diminished in index 5. We speculate that in stage SFEN-4 corresponding to index 5, the semantic information extracted by the convolutional networks is already consistent with the changed area. Therefore, introducing a BSI module at this stage could potentially introduce redundant information, compromising performance in change detection.

To facilitate the comprehension of feature attention regions after integrating the BSI module, Fig. 8 presents a heatmap for feature visualization. As the BSI module is progressively integrated from SFEN-1 to SFEN-4 stages, the region of interest gradually expands from intricate details to encompass the entire scene. By integrating the BSI module across these stages, the model is better equipped to capture and understand the nuances among bitemporal features during the feature learning process. This ensures consistency between the model's focus regions and corresponding labels as shown in rows (III) and (VI). In contrast to the scenario without the BSI module [shown in rows (I) and (IV)], incorporating the BSI module at each stage of the SFEN enables our model to acquire more comprehensive and enriched feature information, as demonstrated in rows (II) and (V). Qualitative analysis validates the effectiveness of the proposed BSI module, particularly highlighting its significant performance improvement when integrated at various feature extraction stages.

2) Effect of the Semantic Categories Number on the SFD-Loss: The number of semantic categories represented in the feature channels, denoted as k plays a crucial role in our proposed SFD-Loss. The SFD-Loss encourages features from different categories across a specified number of channels to form a diverse set, preventing focus on a single highly discriminative region. We hypothesize that the model achieves optimal performance when k is set to 2. This setting divides the feature channels into two categories, aligning the images before and after changes without temporal domain bias among the feature channels. This facilitates the imposition of constraints on the features for optimal feature representation. By imposing

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TABLE VI
RESULTS OF OUR PROPOSED SFD-LOSS FOR DIFFERENT SEMANTIC CATEGORIES k ON LEVIR-CD DATASET

k	1	2	4	8	16
F1 (%)	89.15	**89.27**	89.04	85.38	72.96
IoU (%)	80.43	**80.62**	80.24	74.48	57.43
OA (%)	98.91	**98.93**	98.90	98.56	97.38

feature constraints between corresponding category channels, the model strives to learn discriminative features across different spatial positions. As shown in Table VI, when the number of constrained semantic classes is too large, such as when k is 8 or 16, the number of constrained channels of features becomes more limited, restricting the learning of diverse features. In contrast, while a reduced focus on channels might lead to misjudgments in the final region when k is too small (e.g., $k = 1$), the overall effect remains significantly better than with a larger number of semantic classes. Experimental results confirm that optimal performance is achieved when k is set to 2, which aligns with our hypothesis.

V. DISCUSSION

A. Limited Analysis

Our proposed SFBI-Net outperforms the other nine algorithms on three remote sensing change detection datasets, demonstrating excellent performance. This advantage mainly be attributed to the global interaction of bitemporal features, enhancing the model's capability to perceive changes by obtaining bitemporal feature spatiotemporal relationships. Additionally, by constraining the fused bitemporal features, we maximize the representation of the fused feature. However, our method has limitations, as it still encounters inaccuracies in detection when multiple semantic types of changes are present in the image pairs. These inaccuracies arise from the current model's insufficient understanding of the semantic information associated with these multiple semantic types of changes. We need to further optimize the model's performance under these extreme conditions to ensure stability across various scenarios. Additionally, the design of lightweight bitemporal feature interactions is another aspect that requires further improvement.

Recently, visual-language representation learning [49], [50], [51] has become a popular research topic in computer vision, focusing on utilizing deep learning models to learn representations from image–text pairs. For instance, CLIP [49] proposed to enhance the model's semantic comprehension of images with textual cues through self-supervised learning. In RSICD, ChangeCLIP [52] has introduced a multimodal remote sensing change detection framework that leverages additional context and semantic information from the text to enhance detection performance. Inspired by prior research, our future work aims to enhance our model's performance in scenarios with multiple types of changes by leveraging the semantic information from text prompts.

B. Analysis of Model Evaluation Metric

In RSICD, OA is a common metric used to evaluate the model's ability to detect changes and unchanged areas in pairs of remote sensing images. However, significant limitations of this metric have been identified in extreme scenarios. Specifically, when the model erroneously classifies all image pixels as unchanged despite failing to detect actual changes, the OA metric remains relatively high because most pairs of image regions remain unchanged in the dataset. This observation is further supported by the statistical analysis of extreme cases across three benchmark remote sensing change detection datasets, as shown in Table VII. We suggest introducing other suitable metrics or making appropriate adjustments to the OA metric to comprehensively evaluate the model's overall performance in change detection tasks. For instance, considering the proportions of unchanged and changed regions and applying a weighted average to the final OA metric could be beneficial. This approach balances the influence of different areas on the OA metric and accurately reflects the model's overall detection capability.

TABLE VII
METRIC INDICATOR STATISTICS ON THE EXTREME CASE (SUBSCRIPT 0 AND 1 FOR DISTINGUISHING UNCHANGED AND CHANGED CATEGORIES, RESPECTIVELY)

Dataset	$F1_1(\%)$	$IoU_1(\%)$	$F1_0(\%)$	$IoU_0(\%)$	OA(%)
LEVIR-CD	0.0	0.0	96.70	93.62	93.62
EGY-BCD	0.0	0.0	95.69	91.74	91.74
CLCD	0.0	0.0	95.74	91.83	91.83

VI. CONCLUSION

Detecting changes in remote sensing images remains challenging, primarily due to the complex change patterns in bitemporal images, which include multiple change regions, single large target changes, and small target changes. Based on this, we propose an SFBI-Net for remote sensing change detection. By fusing the BSI model and the SFD-Loss, the performance of our method can be improved. It is shown that the BSI module effectively captures bitemporal feature differences by establishing their spatiotemporal relationships through global interaction, thereby improving the performance of our network, particularly in multiple change regions and single large target changes. The SFD-Loss maximizes the fused bitemporal features in the spatial dimension and further enhances the representation of features in change areas, especially small target areas. Experiments on three benchmark remote sensing change detection datasets demonstrated that our approach outperforms other state-of-the-art methods. It can also be found from the experiments that integrating the BSI module and SFD-Loss improves the model's accuracy in detecting change area boundaries, further enhancing the overall change detection performance.

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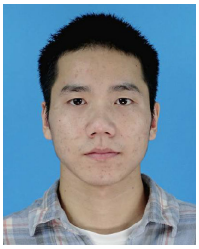
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